From farm gate to fairer markets

KrishiSetu helps Karnataka farmers discover mandi prices, nearby APMCs and buyers—using Kannada or English voice.

1 Prices scattered across sources

Hard to know which price is current and credible

2 Nearby markets and contacts hard to discover

Searching takes time, calls, and guesswork

3 Transport costs make headline prices misleading

High trips eat into earnings



FARMER

PRICE INTELLIGENCE

Observed prices • Market options
Transport cost • Net estimate

→



MARKET / BUYER

Know the observed price.
Compare markets.
Estimate net earnings.

Working prototype
Karnataka-focused



Prices are government/source observations and baseline predictors are anot

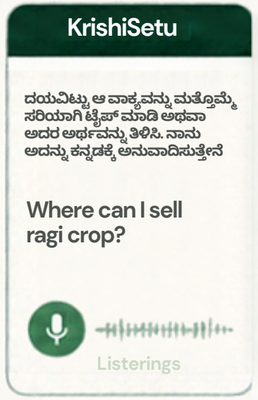
One farmer question.
A complete market decision.

Ask by voice or search

Kannada + English

Gnani Prisma v2.5
speech-to-text

Timbre v2.5
text-to-speech



Find price observations

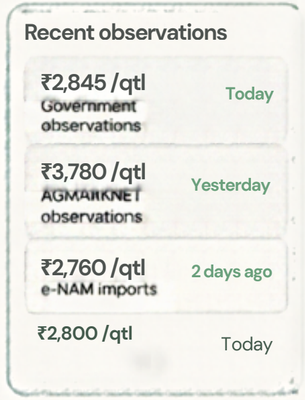
Government

AGMARKNET

e-NAM CSV imports

Internal MongoDB records

Coverage depends on data availability



Compare local options

APMC directory

41 mandi locations

Buyer discovery

Map view



Estimate take-home value

Baseline price estimate

Weather context

Transport-cost calculation

Baseline price (last observed)	₹2,890 /qtl
Weather context	Normal
Transport cost (round trip)	₹320 /qtl
Estimated net	₹2,480 /qtl

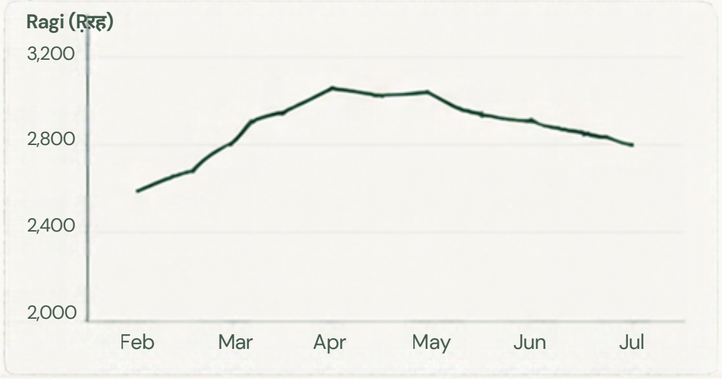
FARMER JOURNEY

Select crop → Compare market observations → Add transport cost → Review estimated net earnings

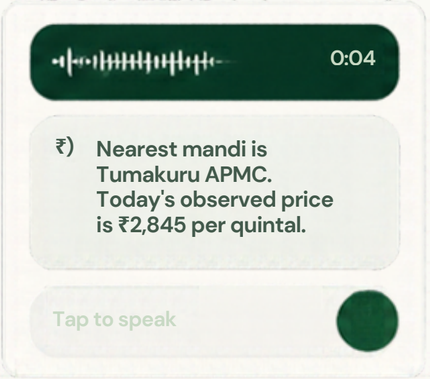
Mandi map view (41 locations)



Price trend (illustrative / based on available observations)



Voice interaction



How we calculate estimated net

Estimated net = expected sale value − transport cost.

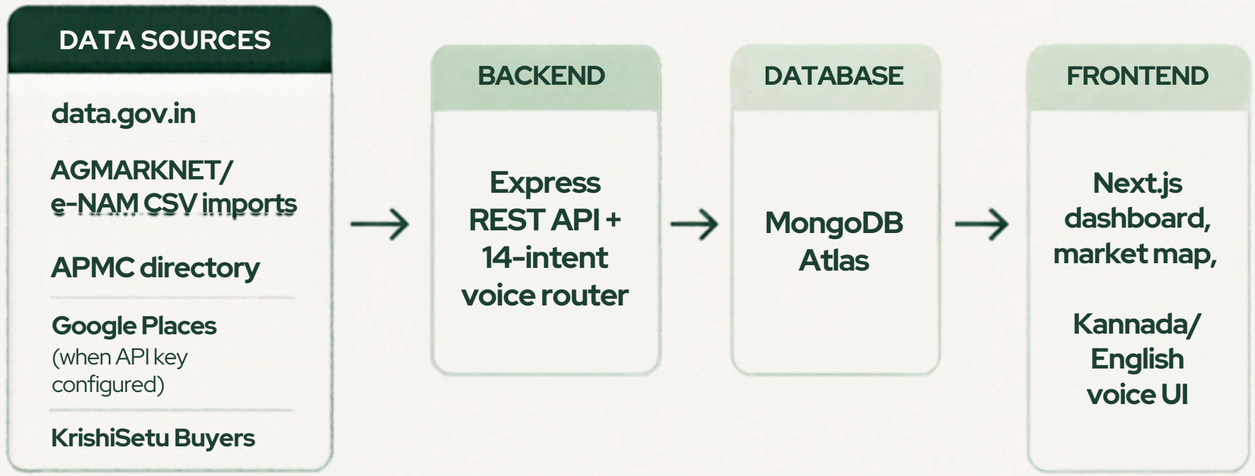
Implemented in prototype	Voice, search, mandi map, observations, calculations
Data/API-dependent	Coverage varies by source and availability

Notes: Baseline forecast uses last-observed-price logic, needs sufficient history, and does not yet model seasonality.

Prices are government/source observations and baseline predictions are not guaranteed market prices.

Built to turn scattered market data into an actionable decision.

PLATFORM ARCHITECTURE



IMPACT AT A GLANCE

16,307 price records	**41** mandi locations	**200+** commodities	**14** intent types
25+ REST endpoints	**20+** React components	**5** source adapters	**10** APMC contact records

TECHNOLOGY STACK

Next.js 16	React 19	TypeScript	Express 5
MongoDB Atlas/ Mongoese	Leaflet + OpenStreetMap/ MapTiler	Open-Meteo	Gnani Prisma v2.5 + Timbre v2.5
OpenRouter optional			

IMPACT FOR FARMERS

Less time searching

clearer market comparisons

more informed sale planning

WHAT'S NEXT

- Expand APMC coverage
- Configure and test live Google Places integration
- Improve historical data and forecasting capabilities

KrishiSetu — Better information. Better-informed selling.

Prices are government/source observations and baseline predictions are not guaranteed market prices.